Cantera vs NN | Test IC 44 | NRMSE (util.py)

T0 = 1031.1 K P0 = 2.272 atm phi0 = 0.770 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	0.231	0.688	2.26	0.999	0.481	5.74	0.363	1.11	1.23	0	Complete
2 P-only DeepONet10	**10k**	0.252	0.701	8.52	0.997	0.528	5.19	0.335	1.82	1.38	0	Complete
2 P-only DeepONet10	**long**	0.186	0.813	0.878	0.4	0.559	0.324	0.312	1.03	1.03	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	0.194	0.773	1.77	1.02	0.628	5.32	0.317	1.03	0.986	0	Complete
3 ANN10 + P-only DeepONet10	**long**	0.174	0.763	1.02	0.379	0.524	0.306	0.292	1.07	1.08	0	Complete
4 T/P DeepONet10	**10k**	0.169	0.784	4.31	0.488	0.547	0.67	0.288	10.8	819	0	Complete
4 T/P DeepONet10	**long**	0.377	0.9	17.3	0.496	0.8	8.83	0.419	1.08	1.01	0	Complete
5 ANN10 + T/P DeepONet10	**10k**	0.782	0.972	57.9	0.965	1	0.638	0.733	221	3.74e+03	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.377	0.871	15	0.721	0.821	9.83	0.432	1.1	1.03	0	Complete
6 T/P/ϕ DeepONet10	**10k**	0.599	2.6	1	1	1.79	1	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	**long**	0.168	0.731	0.687	0.447	0.503	0.276	0.281	1.1	1.19	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	0.599	2.6	1	1	1.79	1	1	0.999	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.149	0.654	0.729	0.423	0.452	0.311	0.251	1.08	1.24	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.